

Defying theories, again

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It seems like in the world of space science and technology, there's a trend going on where everyone's trying to challenge pre-existing theories. A team of researchers led

by Shubham Kanodia from the Carnegie Institution for Science has discovered a planetary system consisting of a giant gas planet that is orbiting a small star dubbed TOI-5205, with the planet being

named TOI-5205b. The star that is being orbited by the newly found planet is categorised as an M dwarf. These stars are known to have a red colour because of their smaller size as compared to the Sun and the fact that they are also about half as hot as the Sun.

The most surprising fact about this discovery, as pointed out by Kanodia in his statement published by carnegiescience.edu, is that the star, which is about four times bigger than Jupiter, our very own gas giant, was able to produce a planet that is as big as Jupiter. He said, "The host star, TOI-5205, is just about four times the size of Jupiter, yet it has somehow managed to form a Jupiter-sized planet, which is quite surprising!"

Kanodia and his team of researchers observed three transits of the planet system using the Astrophysical Research Consortium (ARC) Telescope Imaging Camera at the ARC 3.5 m Telescope at Apache Point Observatory (APO). These transits spanned over a four-month period, on April 22, July 3, and July 16 last year. The team then investigated the data gathered in their readings and established their theories.

As per the information published in the research paper – TOI-5205b: A Short-period Jovian Planet Transiting a Mid-M Dwarf, when the planet crosses its parent star, it blocks about seven per cent of its light. This is one of the 'largest known exoplanet transits'.

Using this information and the exploratory capabilities of the James Webb Telescope, the researchers could further investigate planet formation processes. According to the current theories, the formation of this planet around such a small star was very unlikely. However, the discovery of this planet opens up avenues for research that could not have been thought of earlier.

You can read the research paper here: <https://digit.in/5205b>

